

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : OECAI801C/OECAIDS801C/OECAIML 801C Software Engineering

UPID : 008424

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) What is Feasibility Analysis?
- (II) What is Top-Down Design?
- (III) What is Unit Testing?
- (IV) What is Software Configuration Management (SCM)?
- (V) What is a Dynamic Model?
- (VI) Define System Development Life Cycle (SDLC).
- (VII) What is Problem Partitioning?
- (VIII) What is Information Hiding?
- (IX) What is Quality Assurance (QA)?
- (X) What is an Activity Diagram?
- (XI) Which SDLC model is best suited for high-risk projects?
- (XII) What is the key difference between Functional and Object-Oriented approach?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Explain Static and Dynamic Models with examples. [5]
3. Describe the System Development Life Cycle (SDLC) phases. [5]
4. Explain Top-Down and Bottom-Up design approaches with advantages and disadvantages. [5]
5. Explain Information Hiding and Code Reuse in software engineering. [5]
6. Explain Levels of Testing and Integration Testing with suitable examples. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. Explain System Design in detail along with its principles and objectives. [15]
8. Explain Structured Programming and Object-Oriented Programming in detail with comparison. [15]
9. Explain Staffing and Software Configuration Management (SCM) in detail. [15]
10. Explain the importance of modeling and describe Class Diagram and Collaboration Diagram in detail. [15]
11. Explain the COCOMO Model in detail and demonstrate how it is used for cost and effort estimation. [15]

*** END OF PAPER ***